

Title

Efficacy and safety of a cosmetic anti-acne system versus prescription therapy in adults with facial acne

Abstract

Background: Acne treatment options include prescription and non-prescription formulations with varying tolerability profiles. We propose a comparative effectiveness trial of a novel cosmetic anti-acne regimen versus established prescription therapy. Methods: Adults aged 18-45 with mild-to-moderate facial acne (n=54) will be randomly assigned to either cosmetic salicylic acid regimen (1% cleanser plus 2% treatment lotion, twice daily) or prescription retinoid-benzoyl peroxide combination (once daily), applied to the entire face for 12 weeks. Randomization will be stratified by skin type (Fitzpatrick I-III versus IV-VI). Assessment timepoints: 48 hours, 4 weeks, 8 weeks, and 12 weeks. Primary outcome: clinician-assessed acne severity score. Secondary outcomes: total lesion count, post-inflammatory hyperpigmentation/erythema, sebum production, and skin irritation scores. Tolerability will be measured via validated scales for dryness, redness, and stinging sensations. Inclusion: active facial acne without recent systemic acne medications. Blinding: investigator and participant blinded to treatment allocation. Study design: single-center, double-blind, randomized, active-controlled trial.